VICTORY 2027

FORCE WORK POWER & ENERGY

LECTURE 08 | PHYSICS

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Topics To Be Covered

1. REVISION
2. PRINCIPLE OF CONSERVATION OF ENERGY
3. QUESTIONS
4.

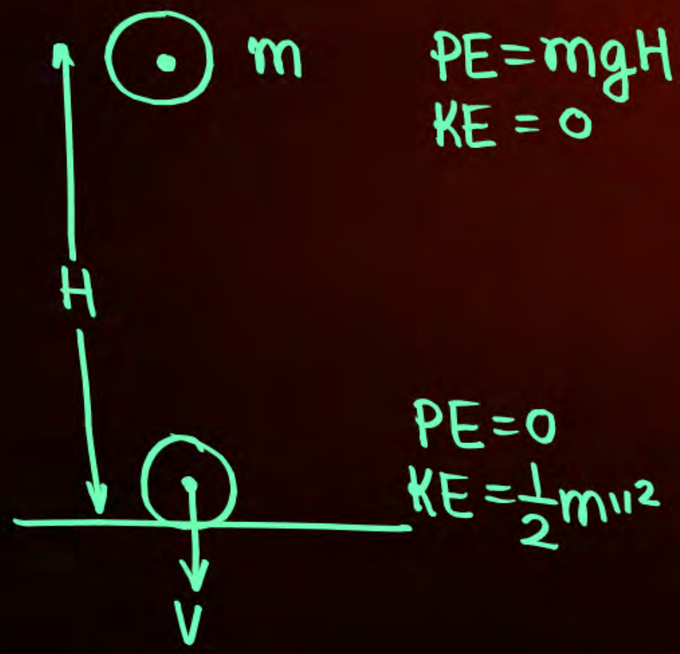

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Physics Wallah

Work-Energy Theorem

$W = K_f - K_i$

$Fd\cos\theta = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

Conservation of energy



$PE = mgH$

$KE = 0$

$PE = 0$

$KE = \frac{1}{2}mv^2$



Elastic Potential energy of spring

↓

KE of block

QUESTIONS

The energy change during photosynthesis in plants is [SEM-1 2021]

1. heat to chemical
2. light to chemical
3. chemical to light
4. chemical to heat

QUESTIONS

In a photovoltaic cell, _____energy converts to _____energy. [SQP 2017]

1. light, wind
2. thermal, kinetic
3. light, electrical
4. heat, chemical

PRINCIPLE OF CONSERVATION OF ENERGY

According to the principle of conservation of energy, energy can neither be created nor can it be destroyed. It only changes from one form to another.

- In universe, energy occurs in various forms. The sum of all forms of energy in the universe remains constant.
- When there is a transformation of energy from one form to another, the total energy always remains same i.e., it remains conserved. (Ideal Case)
- The principle of conservation of energy is one of the fundamental principles of nature.

THEORETICAL VERIFICATION OF K+U = CONSTANT FOR A FREELY FALLING BODY



Initial Position

$KE = 0$

$PE = mgH$

$ME = KE + PE = mgH$

$PE = mg(H-h)$

$KE = \frac{1}{2}mv_i^2 = \frac{1}{2}m(2gh) = mgh$

$ME = KE + PE = mgh + mg(H-h)$

$= mgh + mgH - mgh$

$ME = mgH$

Final Position (Earth)

$V^2 = u^2 + 2gH$

$V^2 = 0 + 2gH$

$PE = 0$

$KE = \frac{1}{2}mv^2$

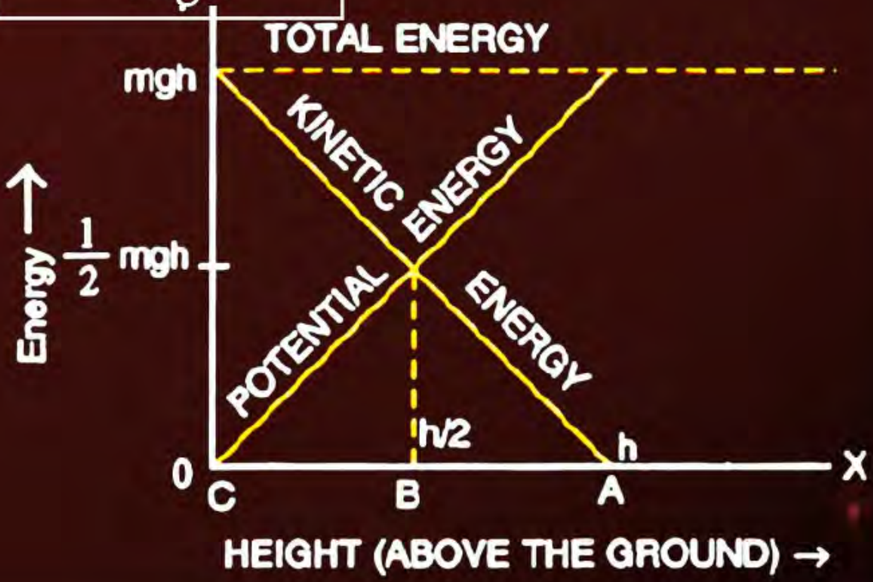
$= \frac{1}{2}m(2gH)$

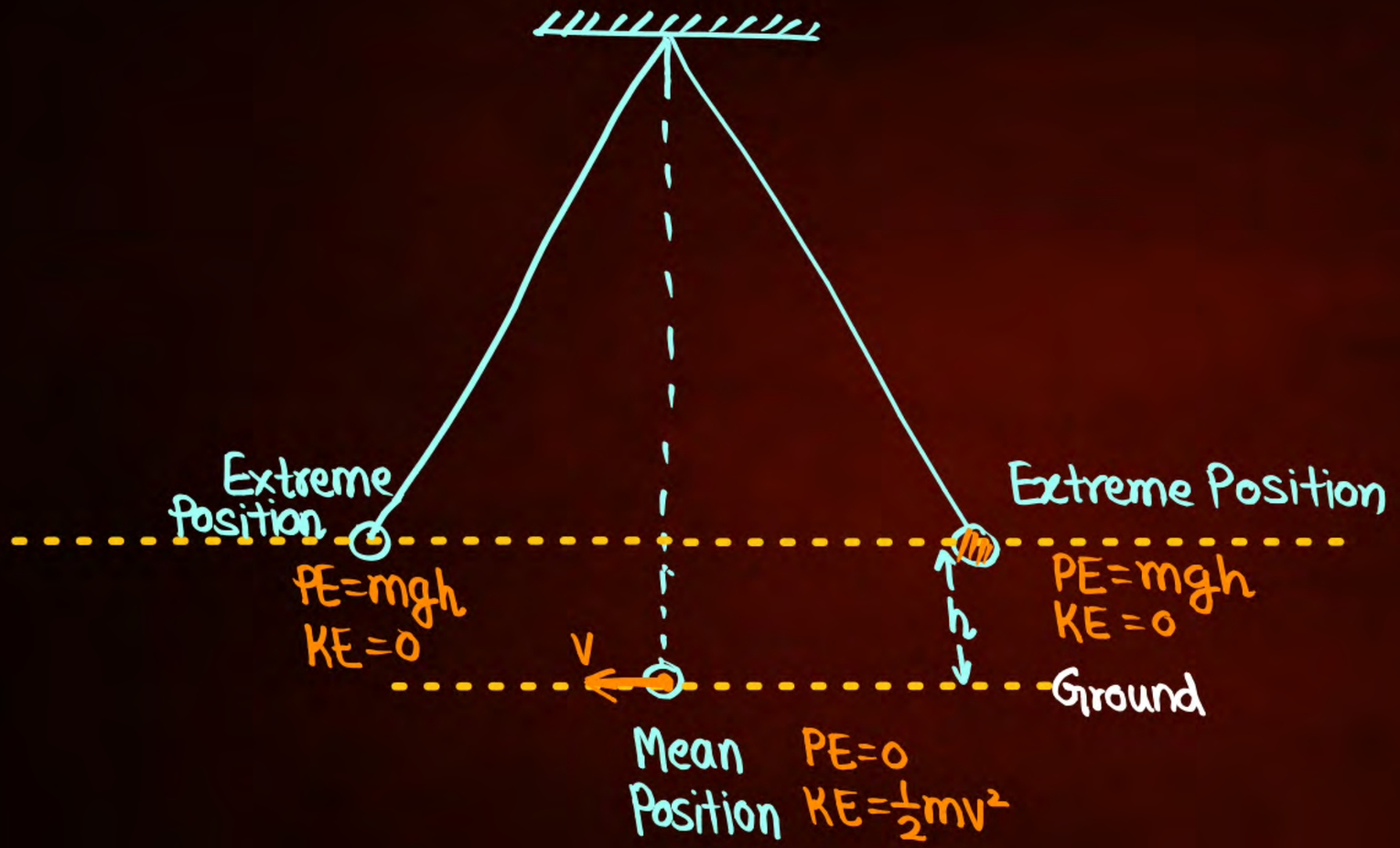
$= mgH$

$ME = KE + PE = mgH$

THEORETICAL VERIFICATION OF K + U = CONSTANT FOR A FREELY FALLING BODY

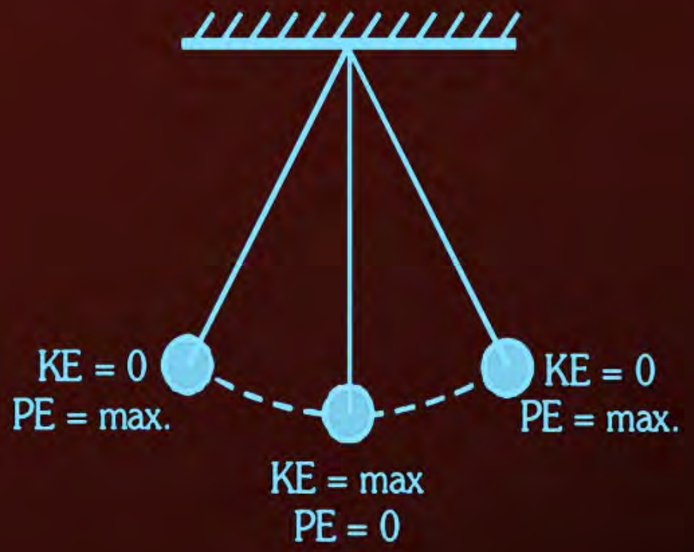
Motion	Height above the ground	Kinetic Energy K	Potential Energy U	Total Energy E = K + U
Downward motion (i.e., free fall)	h (i.e., highest point A)	0	mgh	mgh
	$\frac{1}{2}h$ (i.e., middle point B)	$\frac{1}{2}mgh$	$\frac{1}{2}mgh$	mgh
	0 (i.e., ground C)	mgh	0	mgh
	0 (i.e., ground C)	mgh	0	mgh
Upward motion	$\frac{1}{2}h$ (i.e., middle point B)	$\frac{1}{2}mgh$	$\frac{1}{2}mgh$	mgh
	h (i.e., middle point A)	0	mgh	mgh




Extreme Position
PE=mgh
KE=0
V
Mean Position
PE=0
KE=½mv²
h
Extreme Position
PE=mgh
KE=0
Ground

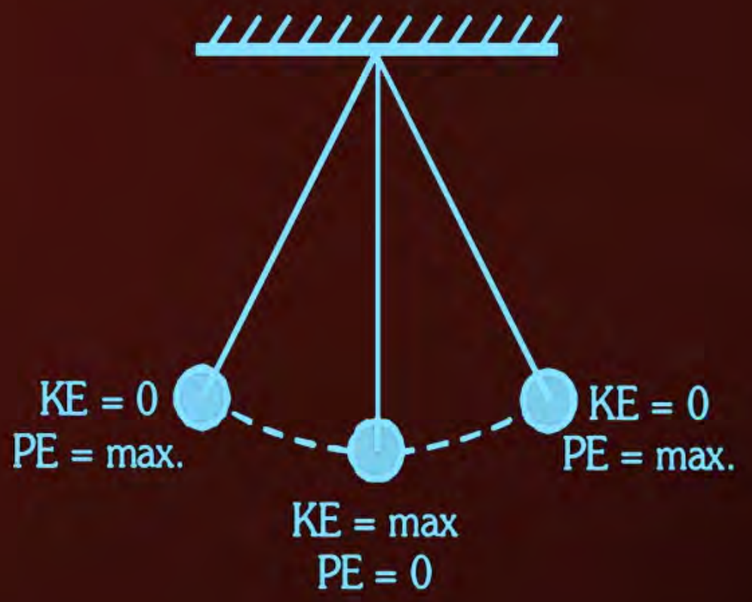
APPLICATION OF PRINCIPLE OF CONSERVATION OF ENERGY TO A SIMPLE PENDULUM

- When the pendulum bob is pulled (say towards left), the energy supplied is stored in it in the form of potential energy (PE) on account of its higher position.
- When the pendulum is released so that it starts moving towards right, then its PE changes into kinetic energy (KE), such that in mean position, it has maximum KE, and zero PE.
- As the pendulum moves towards extreme right, its KE changes into PE; such that at the extreme position, it has maximum PE and zero KE.
- When it moves from this extreme position to mean position, its PE again changes to KE. This means total mechanical energy remains constant.



APPLICATION OF PRINCIPLE OF CONSERVATION OF ENERGY TO A SIMPLE PENDULUM

- This illustrates the law of conservation of energy. Eventually, the bob comes to rest, because during each oscillation a part of the energy possessed by it is transferred to air and in overcoming friction at the point of suspension.
- Thus, the energy of the pendulum is dissipated in air. The law of conservation of energy is not violated because the energy merely changes its form and is not destroyed.

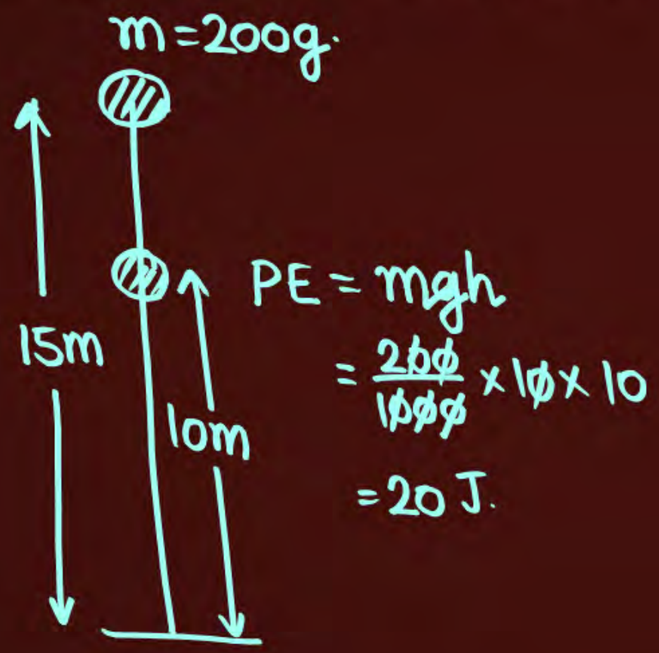


QUESTIONS

A body of mass 200 g falls freely from a height of 15 m . (Take, $g = 10\text{ms}^2$)

(i) When the body reaches 10 m above the ground, its potential energy will be [SEM-1 2021]

1. 20000 J
2. 10 J
3. 10000 J
4. 20 J



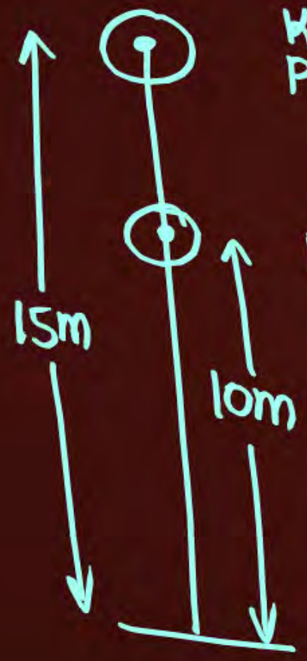
QUESTIONS

A body of mass 200 g falls freely from a height of 15 m . (Take, $g = 10 ms^2$)

(ii) The gain in kinetic energy of the body when it reaches 10 m above the ground is [SEM-1 2021]

loss in PE = Gain in KE

1. 20 J
2. 10 J
3. 30 J
4. 25 J



KE = 0

$PE = mgH = \frac{200}{1000} \times 10 \times 15 = 30\ J.$

$PE = mgh = 20\ J$

QUESTIONS

A body of mass 200 g falls freely from a height of 15 m . (Take, $g = 10 ms^2$)

(iii) The total mechanical energy it will possess, when it is just about to strike the ground is

[SEM-1 2021]

$ME_i = ME_f = 30 J$

1. 30000 J
2. 20000 J
3. 30 J
4. 20 J

QUESTIONS

A body of mass 200 g falls freely from a height of 15 m . (Take, $g = 10\text{ms}^2$)

(iv) The velocity in m/s with which the body will hit the ground is [SEM-1 2021]

1. 30
2. 10
3. $10\sqrt{3}$
4. $10\sqrt{2}$

Initial ME = Final ME

$KE_i + PE_i = KE_f + PE_f$

$0 + 30 = \frac{1}{2}mv^2 + 0$

$30 = \frac{1}{2}\left(\frac{200}{1000}\right)v^2$

$300 = v^2$

$v = \sqrt{300} = 10\sqrt{3} = 17.32\ m/s.$

QUESTIONS

A ball of mass $\mathbf{50g}$ is thrown vertically upwards with an initial velocity $\mathbf{20ms^{-1}}$. Calculate:

(i) the initial kinetic energy imparted to the ball, $KE_i = \frac{1}{2}mu^2 = \frac{1}{2}\frac{50}{1000} \times 20^2 = 10\,J$.

(ii) the maximum height reached if air friction is neglected, and

(iii) the maximum height reached if $\mathbf{40\%}$ of the initial energy is lost against the air friction. Take $g = \mathbf{10ms^{-2}}$.

(ii)



$\frac{1}{2}mu^2 = mgh$

$\frac{1}{2}(20)^2 = 10h$

$h = 20\,m$

(iii) Remaining Energy = 60% of 10

$= 6\,J$

$\frac{1}{2}mu^2 = mgh'$

$6 = \frac{50}{1000} \times 10\, h'$

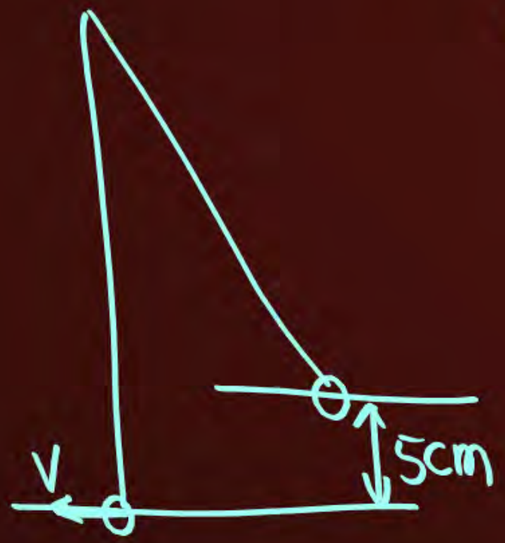
$h' = \frac{60}{5} = 12\,m$

QUESTIONS

A simple pendulum, while oscillating, rises to a maximum vertical height of $\mathbf{5\,cm}$ from its rest position when it reaches to its extreme position on one side. If mass of the bob of simple pendulum is $\mathbf{500g}$ and $g = \mathbf{10ms^{-2}}$, find:

(i) the total energy of simple pendulum at any instant while oscillating, and

(ii) the velocity of bob at its mean position. State the assumption made in your calculation.



(i) $ME_i = KE + PE$

$= 0 + mgh$

$= \frac{500}{1000} \times 10 \times \frac{5}{100}$

$= \frac{1}{4}$ J.

(ii) $ME_i = ME_f$

$\frac{1}{4} = \frac{1}{2} mv^2$

$\frac{1}{4} = \frac{1}{2}\left(\frac{500}{1000}\right) v^2$

$v = 1$ m/s

HOMEWORK

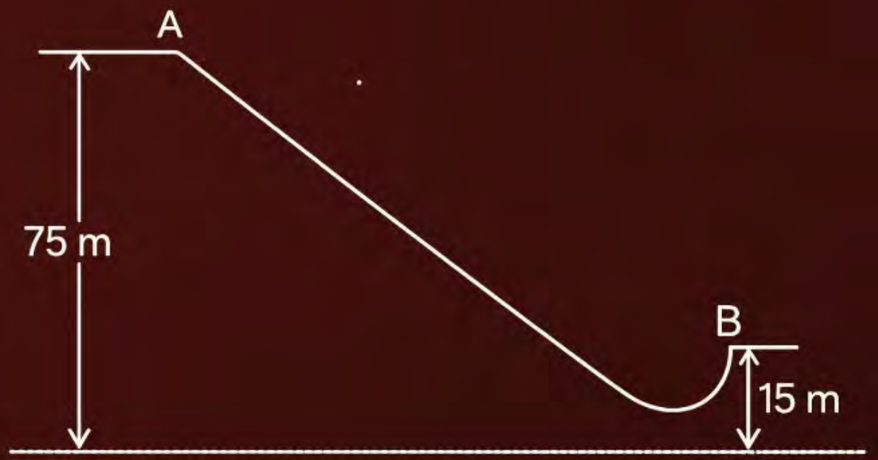
- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

The diagram given below shows a ski jump. A skier weighing 60kgf stands at A at the top of ski jump. He moves from A and takes off for his jump at B.

(a) Calculate the change in the gravitational potential energy of the skier between A and B.

(b) If 75% of the energy in part (a) becomes kinetic energy at B, calculate the speed at which the skier arrives at B. (Take $g = 10\text{ms}^{-2}$)



Thank You